

Taner Giray Sönmez

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EDUCATION

- Columbia University** New York City, USA
Master of Science in Machine Learning - Computer Science; GPA: 3.95/4 September 2024 - December 2025
Related Coursework: High-Performance Machine Learning, Machine Learning Cloud, Neural Networks and Deep Learning, Applied Machine Learning, Natural Language Processing, Artificial Intelligence
- Sabanci University** Istanbul, Turkey
Bachelor of Science - Computer Science; GPA: 3.82/4 September 2019 - June 2024
Honors and Awards: Dean's List - High Honor : 6x (Six Terms), 50% Encouragement Scholarship (Aug 2022 and Aug 2023)
Related Coursework: Advanced Programming, Data Science, Machine Learning, Data Analysis and Research in Business, Mobile Application Development, Software Engineering, Computer Networks, Database Systems, Optimization Models in Business Analytics
- Uppsala University** Uppsala, Sweden
Bachelor of Science - Computer Science; Erasmus Exchange Program January 2023 - June 2023
Honors and Awards: Monthly Erasmus Grant

INDUSTRY EXPERIENCE

- Amazon Web Services (AWS)** New York City, USA
Software Development Engineer Intern - AWS Glue May 2025 - August 2025
 - : Enhanced Personally Identifiable Information (PII) name detection in AWS Glue by deploying a new ML model, reducing false negatives by 37% and false positives by 20%.
 - : Accelerated data processing speed for customers, improving efficiency for common jobs on smaller datasets.
 - : Expanded PII detection to include 15+ new entity types, like organizations and locations, at no additional computational cost.
 - : Researched over 12 machine learning models to select and implement the optimal one for its superior performance.
- Mars Athletic Club (MACFIT)** Istanbul, Turkey
Data Scientist (Part-Time) July 2023 - June 2024 and July 2022 - January 2023
 - : Developed a 'Churn Prediction Model' using statistical analysis and machine learning algorithms, specifically employing boosting methods such as XGBoost. Preprocessed a dataset with millions of rows of customer daily visit records. Improved contract renewal rate, generating over \$45,000 in additional monthly revenue.
 - : Conducted clustering with unsupervised learning to segment customers and identify their unique characteristics for more accurate predictions.
 - : Utilized SQL to extract, manipulate, and preprocess data for modeling purposes as well as for daily tasks.
 - : Implemented sentiment analysis on customer comments using NLP models, such as BERT, to categorize feedback by key areas and sentiment, identifying trends in satisfaction and opportunities for improvement.

RESEARCH EXPERIENCE

- Columbia University - Internet Real Time Lab** New York City, USA
AI Researcher January 2025 - May 2025
 - : Engineered an LLM-based system for privacy policy analysis, achieving 99% weighted recall in clause classification by integrating a Retrieval-Augmented Generation (RAG) pipeline with a Pinecone vector database.
 - : Created and open-sourced large annotated privacy policy datasets, comprising over 5,500 clauses from 450+ companies, to foster research in automated policy analysis.
 - : Pioneered DSPy pipelines to auto-optimize prompts and fine-tune lightweight models, boosting performance of a GPT-4o mini model to a 0.98 F1 score.
 - : Contributed to a proof-of-concept Chrome extension that leverages LLMs to provide real-time privacy risk alerts, investigating caching mechanisms and fallback strategies to ensure system reliability.
- Mars Athletic Club - Sabanci University** Istanbul, Turkey
ML Researcher September 2023 - July 2024
 - : Studied exercise habit formation using machine learning and causal inference on data from Mars Athletic, Turkey's largest gym chain, supervised by Dr. Onur Varol.
 - : Collaborated with 3 peers to develop a novel survival metric for assessing exercise habit durability.
 - : Analyzed large-scale dataset from 200,000+ gym members to identify key factors influencing habit formation and provide actionable insights for gym management and public health.
 - : First-authored the final project report detailing the study's methods and results; a journal publication is in progress.
 - : ArXiv link can be accessed from here.

TEACHING EXPERIENCE

- Sabanci University** Istanbul, Turkey
Learning Assistant September 2021 - June 2022
 - : Worked as a Learning Assistant for the 'Computational Approaches to Problem Solving' course, teaching Python and problem solving to freshman students.
 - : Reinforced lecture topics and provided assistance to students during weekly recitation hours.

PROJECTS

- **LightGPT2: Efficient GPT-2 Optimization**
Machine Learning Engineer *January 2025 - May 2025*
 - : Developed LightGPT2, a highly compressed and efficient variant of GPT-2, by designing a multi-stage pipeline combining Knowledge Distillation, LoRA, 80% unstructured pruning, and 4-bit quantization.
 - : Achieved a 66% reduction in training memory and a 27.3% reduction in inference memory, enabling fine-tuning and deployment on consumer GPUs with less than 6GB of VRAM.
 - : Maximized inference performance by integrating FlashAttention, resulting in a 3x speedup in forward throughput while maintaining comparable model quality (perplexity increase of only 0.89).
 - : Skills: PyTorch, Hugging Face, Knowledge Distillation, Low-Rank Adaptation (LoRA), Pruning, Quantization, FlashAttention.
- **Neural Document Search and Q&A System**
Machine Learning Engineer *January 2025 - May 2025*
 - : Built and trained a dual-component neural search system with a DistilBERT-based retriever and reader, capable of processing natural language queries and extracting precise answers from a large document corpus.
 - : Fine-tuned the reader model to achieve 86.5% F1 score on the SQuAD v1.1 benchmark, approaching state-of-the-art performance with a lightweight and efficient architecture.
 - : Designed the system for scalable deployment on Google Cloud Platform.
 - : Skills: PyTorch, DistilBERT, Google Cloud Platform.
- **Sarcasm Detection in Text**
Machine Learning Researcher *January 2025 - May 2025*
 - : Conducted a comparative study of transformer models (RoBERTa, DeBERTa) and hierarchical recurrent architectures (HAN, FAN) for sarcasm detection.
 - : Fine-tuned and evaluated models on multiple diverse datasets (iSarcasmEval, Sarcasm Corpus V2), demonstrating that transformers significantly outperform RNN-based approaches.
 - : Validated model generalization on the BigBench SNARKS benchmark, where the fine-tuned DeBERTa model achieved 76% accuracy, showcasing strong cross-domain performance.
 - : Skills: PyTorch, Hugging Face, DeBERTa, RoBERTa.
- **FFP Tracker**
Web Developer *September 2022 - January 2023*
 - : Led a team of 3 in developing the FFP Tracker, a web platform that enhances transparency in the Turkish Football Federation by requiring teams to submit financial documents for review by the federation and its legal team.
 - : Developed forecasting and monitoring tools to prevent breaches and ensure that teams generate revenue within budget, promoting accountability and transparency through public information access.
 - : Skills: Frontend Development, React, PostgreSQL, ExpressJS, NodeJS, Agile Development, Git, Full Stack Development.
- **Socket Programming Quiz Game**
Developer *September 2022 - January 2023*
 - : Developed a client-server application, a quiz game, consisting of two modules: a server and a player.
 - : Implemented the server module to manage connections, pose questions, and track scores. Developed the player module for user enrollment and real-time score updates.
 - : Skills: C#.
- **Ace**
Mobile App Developer *February 2022 - June 2022*
 - : Developed a mobile application, Ace, which serves as a social media platform for sports lovers, using Flutter with a team of 4.
 - : Modeled the app after Instagram and utilized Flutter's widgets to create a unique user interface and user experience.
 - : Skills: Dart, Flutter, Frontend Development, Backend Development, App State Management, Firebase, Git, Android Development.

SKILLS SUMMARY AND CERTIFICATIONS

- **Programming Languages:** Python, C++, SQL/noSQL, Dart, R (introductory experience).
- **Tools and Frameworks:** Scikit-learn, Numpy, Google Cloud Platform, Pandas, Selenium, Large Scale Data Processing, Flutter, React, Git, PostgreSQL, MySQL, PyTorch, MongoDB, Verilog (introductory experience).
- **Soft Skills:** Agile Methodologies, Analytical Thinking, Teamwork, Problem Solving, Effective Communication, Team Player, Adaptability, Project Management.

VOLUNTARY WORK

- **Gazi Mustafa Pasa Primary School**
Teacher *Kocaeli, Turkey
February 2020 - June 2020*
 - : Participated in a volunteer project, visiting 4th-grade students weekly to assist with schoolwork and provide lessons on general culture, making a positive impact on their education and supporting their growth.
 - : Developed interpersonal and communication skills, as well as the ability to work effectively in a team. Reinforced commitment to making a positive impact in the lives of others, showcasing a dedication to community service and social responsibility.